

SETIF AIN ARNAT, Algeria

WMO: 604450

Lat: 36.178N

Lon: 5.324E

Elev: 1050

StdP: 89.33

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.1	-6.4	2.5	7.2	-5.0	2.8	6.2	12.4	5.9	10.9	6.0	2.7	0	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	36.8	17.8	35.3	17.5	34.0	17.3	20.0	30.6	19.2	30.1	18.6	29.5	5.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	13.9	22.8	16.1	13.0	22.0	15.2	12.2	21.8	61.9	30.4	59.3	30.2	57.2	29.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.9	DB	-4.5	39.1	2.1	1.5	-6.0	40.2	-7.3	41.1	-8.4	42.0	-10.0	43.1	(4)
(5)			WB	-5.2	22.1	2.1	1.9	-6.7	23.5	-7.9	24.6	-9.1	25.7	-10.6	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	5.6	6.2	9.3	12.3	17.0	22.4	26.5	25.8	20.8	16.3	10.0	6.5
	DBStd	8.11	2.69	3.36	3.65	3.58	4.29	4.18	2.92	2.96	3.42	3.52	3.19	2.61
	HDD10.0	478	138	113	56	17	2	0	0	0	0	2	39	112
	HDD18.3	2004	396	341	279	184	77	12	0	0	14	83	251	367
	CDD10.0	2285	2	6	35	84	220	373	513	491	323	196	38	3
	CDD18.3	769	0	0	1	2	37	135	254	233	88	19	0	0
	CDH23.3	8948	0	0	3	32	416	1627	3201	2688	796	184	1	0
	CDH26.7	4379	0	0	0	2	122	767	1766	1394	285	43	0	0
Wind	WSAvg	3.6	3.5	3.9	3.9	3.9	3.7	3.7	3.5	3.5	3.4	3.2	3.6	3.3
Precipitation	PrecAvg	384	42	33	35	37	40	25	9	15	38	28	37	43
	PrecMax	466	130	102	102	89	101	73	40	35	120	64	118	133
	PrecMin	299	6	6	0	2	3	0	0	0	1	1	6	0
	PrecStd	54	31	20	25	24	29	21	11	11	30	19	25	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	19.1	23.1	26.1	32.7	36.9	38.8	38.0	33.5	30.1	22.2	17.0
		MCWB	9.2	9.2	11.7	13.5	15.5	16.8	18.2	18.2	17.2	16.1	13.3	9.6
	2%	DB	14.2	16.2	20.2	24.0	29.9	35.0	37.1	36.3	31.8	27.8	19.3	14.5
		MCWB	8.6	9.0	10.6	12.7	15.5	16.9	18.0	17.8	17.2	15.1	11.7	8.9
	5%	DB	12.2	14.1	18.2	21.9	27.8	33.2	35.9	35.0	30.0	25.2	17.4	12.9
		MCWB	7.6	8.2	9.9	12.4	15.1	16.4	17.8	17.7	16.8	14.4	10.8	8.5
	10%	DB	10.5	12.1	16.3	19.6	25.7	31.2	34.3	33.3	28.0	23.0	15.6	11.2
		MCWB	6.7	7.2	9.4	11.7	14.4	16.4	17.4	17.5	16.3	14.0	10.3	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.8	13.1	15.2	17.8	19.8	21.2	21.1	20.3	17.7	14.7	11.1
		MCDB	14.4	16.1	20.1	22.7	27.1	31.5	31.9	31.1	28.7	25.6	19.0	14.7
	2%	WB	9.2	9.8	11.8	14.1	16.8	18.6	19.8	20.0	18.9	16.6	12.9	9.7
		MCDB	13.1	14.3	18.2	21.0	26.0	29.9	31.8	30.1	26.7	23.7	17.3	13.2
	5%	WB	8.1	8.6	10.8	13.2	16.0	17.8	19.1	19.2	18.0	15.7	11.8	8.9
		MCDB	11.5	13.1	17.0	19.8	25.0	28.9	31.7	29.5	25.4	22.3	15.8	12.0
	10%	WB	7.1	7.7	9.8	12.2	15.1	17.1	18.4	18.6	17.3	14.9	10.8	8.1
		MCDB	10.1	11.5	15.5	18.4	24.0	27.8	31.0	29.2	24.6	20.8	14.8	10.8
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.3	10.4	11.2	12.5	13.9	14.6	13.8	12.1	10.9	9.3	8.5
		MCDBR	12.3	12.9	13.8	14.8	15.1	15.3	15.5	15.2	14.3	14.3	12.3	11.3
		MCWBR	7.8	7.3	6.8	6.8	5.8	5.0	5.0	5.0	4.7	5.7	6.6	7.1
	5% WB	MCDBR	11.3	11.4	13.1	13.3	14.0	15.3	15.1	14.3	12.6	11.9	11.2	9.9
		MCWBR	7.5	7.0	7.1	7.0	6.0	5.8	5.5	5.3	4.7	5.2	6.6	6.6
Clear-Sky Solar Irradiance	taub	0.292	0.313	0.369	0.403	0.439	0.440	0.487	0.457	0.427	0.378	0.332	0.300	
	taud	2.506	2.440	2.260	2.178	2.093	2.107	1.965	2.053	2.154	2.283	2.415	2.500	
	Ebn at Noon	912	928	895	880	849	845	802	819	824	836	849	876	
	Edh at Noon	80	97	127	146	161	159	182	163	140	112	88	77	
All-Sky Solar Radiation	RadAvg	2.49	3.26	4.51	5.59	6.53	7.51	7.55	6.67	5.13	3.90	2.71	2.23	
	RadStd	0.26	0.32	0.38	0.43	0.44	0.26	0.18	0.22	0.25	0.25	0.18	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page